Junjie (Takku) Li

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EDUCATION

Northeastern University **Sep 2023 - Dec 2025**
M.S., Computer Science *Seattle, WA*

- **GPA:** 3.9/4, Teaching Assistant for Object-oriented programming(OOP) and Advanced Algorithm
- **Coursework:** Distributed Systems, Database, Cloud Computing, Computer Vision, Computer System, Operating Systems

Nankai University **Sep 2017 - Jun 2021**
B.S., Mathematics *Tianjin, China*

EXPERIENCE

AI/ML Engineer Intern **Jun 2025 - Aug 2025**
TechX *Redmond, WA*

- Built tokenizer (BPE) and Transformer LM from scratch with **PyTorch**, achieving low perplexity on TinyStories/OpenWebText.
- Optimized training throughput by 3× via **Triton**-based fused kernels, mixed precision, and multi-GPU parallelism.
- Modeled compute-optimal scaling laws and predicted ideal model/data sizes under fixed FLOPs budgets.
- Collaborated full pretraining + alignment pipeline (data cleaning, SFT, RLHF), improving response helpfulness by 25%.
- Automated 70%+ of codebase generation through **Vibe Coding** with **Claude Code** and custom **Cursor** rules, leveraging **prompt engineering** and enforcing **MVVM** architectural consistency to accelerate development while ensuring reliability and scalability.

Machine Learning Engineer Intern **Jun 2024 - Aug 2024**
Berkshire Hathaway HomeServices *Seattle, WA*

- Engineered a prediction system with deep learning models using **PyTorch**, achieving a 30% accuracy improvement.
- Designed **RESTful APIs** with **Flask** for model **inference**, integrating seamlessly with the system to support service interactions.
- Developed performance metric dashboards using Jupyter Notebook with **pandas** and Plotly for interactive data visualization.
- Applied **Agile Scrum** practices to optimize data pipelines, reducing development cycle time by 25% through automation.

PhD student in information system and quantitative marketing, Research Assistant **Sep 2021 - May 2023**
Temple University *Philadelphia, PA*

- Examined the influence of streaming comments on video appreciation through a data-driven research project in **Python, Dask** and **R**.
- Conducted a high-performance **NLP** pipeline by combining fine-tuned **transformer** models for topic classification and sentiment analysis, achieving scalable and accurate text understanding, demonstrating proficiency in advanced **AI/ML** libraries.
- Applied **causal inference** methods (Synthetic Control, Difference-in-Difference) to design a quasi-experiment, revealing significant causal effects on viewer engagement and like rate.

PROJECTS

LLM-Driven TurtleBot

- Developed a ROS 2–based TurtleBot system on **Linux**, integrating **LLM**-driven natural language understanding with Nav2 navigation and real-time motor/sensor firmware.
- Engineered **multi-threaded C++** and **Python** ROS nodes, configured DDS networking for multi-node communication over **TCP/UDP** and interfaced with microcontrollers, showing **system integration**, **concurrency**, and **embedded development** experience.
- Implemented **shell scripting** for deployment and debugging, bridging high-level AI planning with real-time hardware control.

Game Recommendation via RAG with vLLM on MCP | Link

- Developed an **RAG** pipeline that recommends games based on public reviews and player preferences.
- Indexed review embeddings into **FAISS** and retrieved context to augment prompts with real user sentiment.
- Deployed the **LLM** endpoint via **MCP**, enabling scalable and containerized model inference across GPU nodes.
- Served LLM via vLLM for high-throughput generation with **OpenAI**-compatible API and low latency.
- Achieved sub-second latency under concurrent requests using **vLLM's** paged attention and continuous batching.

AI-Driven Flashcard Generator | Qualcomm & Microsoft AI Hackathon Project | Link

- Built AI-powered flashcard generator with **Python** and **FastAPI**, leveraging LLMs for context-awareness within an **MVC** backend.
- Proposed energy-efficient LLM inference pipeline on edge devices, optimizing for privacy and battery consumption.
- Implemented quantized **Llama**-v3.1-8B using **ONNX** and Qualcomm Genie, reducing model size by 70%.
- Performed stateless chunking and **KV cache** strategies, significantly reducing latency and stabilizing memory usage.

TECHNICAL SKILLS

- **Languages & Databases**: Python(5yrs), Triton, Golang, C/C++, Scala, MySQL, Redis, PostgreSQL, MongoDB, FAISS, Redshift
- **Frameworks & Tools**: PyTorch, Flask, Spring Boot, Express, Angular, FastAPI, Django, Dask, Kubernetes, Terraform, Spark, Linux
- **Cloud**: AWS(Certified Solutions Architect Associate), Azure (Certified AI Engineer Associate), CI/CD, GitHub Actions
- **AI & Machine Learning**: Large Language Models, Prompt Engineering, Natural Language Processing, Computer Vision, AI infra, RAG, Reinforcement Learning